

COURSE CODE:ITA0608

COURSE NAME: Machine learning

EXPERIMENT-11: Write a program for the task of Credit Score Classification

Aim: To classify customer credit scores using the **Decision Tree Classifier** in Python.

Algorithm:

1. **Start**
2. Import the required libraries (pandas, train_test_split, DecisionTreeClassifier, accuracy_score).
3. Read the **credit_score.csv** dataset.
4. Store the input attributes in **X** and the target (CreditScore) in **y**.
5. Split the dataset into training data and testing data.
6. Create a **Decision Tree Classifier** model.
7. Train the model using the training dataset.
8. Predict the credit score for the testing dataset.
9. Calculate the accuracy of the model.
10. Display the predicted credit scores and accuracy.
11. Give a new customer's details as input.
12. Predict whether the customer's credit score is **Good** or **Poor**.
13. Display the predicted result.
14. **Stop**.

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

data = pd.read_csv("knn_dataset.csv")

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Values:")
print(list(y_test))

print("\nPredicted Values:")
print(list(y_pred))

accuracy = accuracy_score(y_test, y_pred)
```

Output:

```
... Predicted Credit Scores:
['Good' 'Good' 'Good' 'Good' 'Good' 'Good']

Accuracy:
1.0

Classification Report:
              precision    recall  f1-score   sup
```

Result: The K-Nearest Neighbours (K-NN) algorithm was successfully implemented using Python. The classifier was trained with the given dataset, predicted the class labels for the test data, and achieved good classification accuracy.